

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 1-Chlorooctane

**CAS No** 111-85-3

**Synonyms** 1-Octylchloride

**Product Code** **L04358**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

No hazards identified

#### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 1

#### Label Elements



Environment

**Signal Word****Danger****Hazard Statements**

H410 - Very toxic to aquatic life with long lasting effects  
Combustible liquid

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P235 - Store in a well-ventilated place. Keep cool  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

## Section 3 - Composition and Information on Ingredients

| Component      | CAS No   | Weight % |
|----------------|----------|----------|
| Octyl chloride | 111-85-3 | >95      |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                                                       |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                        |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting     |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                  |

## Section 5 - Fire Fighting Measures

**Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Hazardous Decomposition Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

**Specific Hazards Arising from the Chemical**

Combustible material. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Ensure adequate ventilation. Avoid contact with skin and eyes. Avoid contact with skin and clothing. Avoid breathing vapors or mists. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Wash thoroughly after handling.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties****Appearance**

Colorless

**Physical State**

Liquid

**Odor**

Odorless

**Odor Threshold**

No data available

**pH**

No information available

**Melting Point/Range**

-61 °C / -77.8 °F

**Softening Point**

No data available

**Boiling Point/Range**

183 °C / 361.4 °F

**Flash Point**

68 °C / 154.4 °F

**Method** - No information available

|                                         |                          |                                        |
|-----------------------------------------|--------------------------|----------------------------------------|
| Evaporation Rate                        | No data available        |                                        |
| Flammability (solid,gas)                | Not applicable           | Liquid                                 |
| Explosion Limits                        | No data available        |                                        |
| Vapor Pressure                          | 66 mbar @ 99 °C          |                                        |
| Vapor Density                           | 5.13                     | (Air = 1.0)                            |
| Specific Gravity / Density              | 0.870                    |                                        |
| Bulk Density                            | Not applicable           | Liquid                                 |
| Water Solubility                        | 0.02 g/l                 | practically insoluble                  |
| Solubility in other solvents            | No information available |                                        |
| Partition Coefficient (n-octanol/water) |                          |                                        |
| Component                               | log Pow                  |                                        |
| Octyl chloride                          | 5.42                     |                                        |
| Autoignition Temperature                | 230 °C / 446 °F          |                                        |
| Decomposition Temperature               | No data available        |                                        |
| Viscosity                               | 1.24 mPa.s at 20 °C      |                                        |
| Explosive Properties                    |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available |                                        |
| <b>Other information</b>                |                          |                                        |
| Molecular Formula                       | C8 H17 Cl                |                                        |
| Molecular Weight                        | 148.68                   |                                        |

## Section 10 - Stability and Reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                               |
| Stability                        | Stable under normal conditions.                                                          |
| Conditions to Avoid              | Incompatible products, Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | None known.                                                                              |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Hydrogen chloride gas.          |
| Hazardous Polymerization         | No information available.                                                                |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

|                                        |                                                             |
|----------------------------------------|-------------------------------------------------------------|
| Product Information                    | No acute toxicity information is available for this product |
| (a) acute toxicity;                    |                                                             |
| Oral                                   | No data available                                           |
| Dermal                                 | No data available                                           |
| Inhalation                             | No data available                                           |
| (b) skin corrosion/irritation;         | No data available                                           |
| (c) serious eye damage/irritation;     | No data available                                           |
| (d) respiratory or skin sensitization; |                                                             |
| Respiratory                            | No data available                                           |
| Skin                                   | No data available                                           |
| (e) germ cell mutagenicity;            | No data available                                           |

|                                            |                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                      |
| (g) reproductive toxicity;                 | No data available                                                                                                   |
| (h) STOT-single exposure;                  | No data available                                                                                                   |
| (i) STOT-repeated exposure;                | No data available                                                                                                   |
| Target Organs                              | No information available.                                                                                           |
| (j) aspiration hazard;                     | No data available                                                                                                   |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                      |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |

## Section 12 - Ecological Information

|                                       |                                                                                                                                                                                                                                                                       |                    |                               |                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------|-------------------------------------------------|
| Ecotoxicity effects                   | Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.                                                                                   |                    |                               |                                                 |
| Component                             | Freshwater Fish                                                                                                                                                                                                                                                       | Water Flea         | Freshwater Algae              | Microtox                                        |
| Octyl chloride                        | LC50=12 mg/L 48 h                                                                                                                                                                                                                                                     | EC50=0.21 mg/L 48h |                               | EC50 = 291 mg/L 5 min<br>EC50 = 398 mg/L 30 min |
| Persistence and Degradability         | Not readily biodegradable                                                                                                                                                                                                                                             |                    |                               |                                                 |
| Persistence                           | May persist.                                                                                                                                                                                                                                                          |                    |                               |                                                 |
| Degradation in sewage treatment plant | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                                                                                                                                       |                    |                               |                                                 |
| Bioaccumulative Potential             | Product has a high potential to bioconcentrate                                                                                                                                                                                                                        |                    |                               |                                                 |
| Component                             | log Pow                                                                                                                                                                                                                                                               |                    | Bioconcentration factor (BCF) |                                                 |
| Octyl chloride                        | 5.42                                                                                                                                                                                                                                                                  |                    | No data available             |                                                 |
| Mobility                              | Spillage unlikely to penetrate soil. The product is insoluble and floats on water. Is not likely mobile in the environment due its low water solubility Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles |                    |                               |                                                 |
| Endocrine Disruptor Information       | This product does not contain any known or suspected endocrine disruptors                                                                                                                                                                                             |                    |                               |                                                 |
| Persistent Organic Pollutant          | This product does not contain any known or suspected substance                                                                                                                                                                                                        |                    |                               |                                                 |
| Ozone Depletion Potential             | This product does not contain any known or suspected substance                                                                                                                                                                                                        |                    |                               |                                                 |

## Section 13 - Disposal Considerations

|                                     |                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| Other Information                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                      |

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Hazard Class 9  
Packing Group III

| Component                          | IMDG Marine Pollutant                                 |
|------------------------------------|-------------------------------------------------------|
| Octyl chloride<br>111-85-3 ( >95 ) | IMDG regulated marine pollutant (Listed in the index) |

### ADG

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Hazard Class 9  
Packing Group III

### IATA

UN-No UN3082  
Proper Shipping Name Environmentally hazardous substances, liquid, n.o.s.  
Hazard Class 9  
Packing Group III

Environmental hazards Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### Standard for the Uniform Scheduling of Medicines and Poisons

No poison schedule number allocated.

#### Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component                 | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|---------------------------|-------------------------------------------------------------|------------------------|
| Octyl chloride - 111-85-3 | Present                                                     | -                      |

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component      | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Octyl chloride | X    | X     | 203-915-5 | -      | X    | -   | X    | X     | X    | X    | X     | KE-05795 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**MARPOL** - International Convention for the Prevention of Pollution from Ships

| Component                 | IMDG Marine Pollutant                                 |
|---------------------------|-------------------------------------------------------|
| Octyl chloride - 111-85-3 | IMDG regulated marine pollutant (Listed in the index) |

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                 | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|---------------------------|------------------------------------|------------------------------------------------------------------------|
| Octyl chloride - 111-85-3 | Annex I - Y45                      | Y45 except substances referenced in Annex I                            |

| Component      | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Octyl chloride | 111-85-3 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

## Section 16 - Other Information



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**Legend**

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

|                         |                 |
|-------------------------|-----------------|
| <b>Revision Date</b>    | 18-Nov-2022     |
| <b>Revision Summary</b> | Not applicable. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet